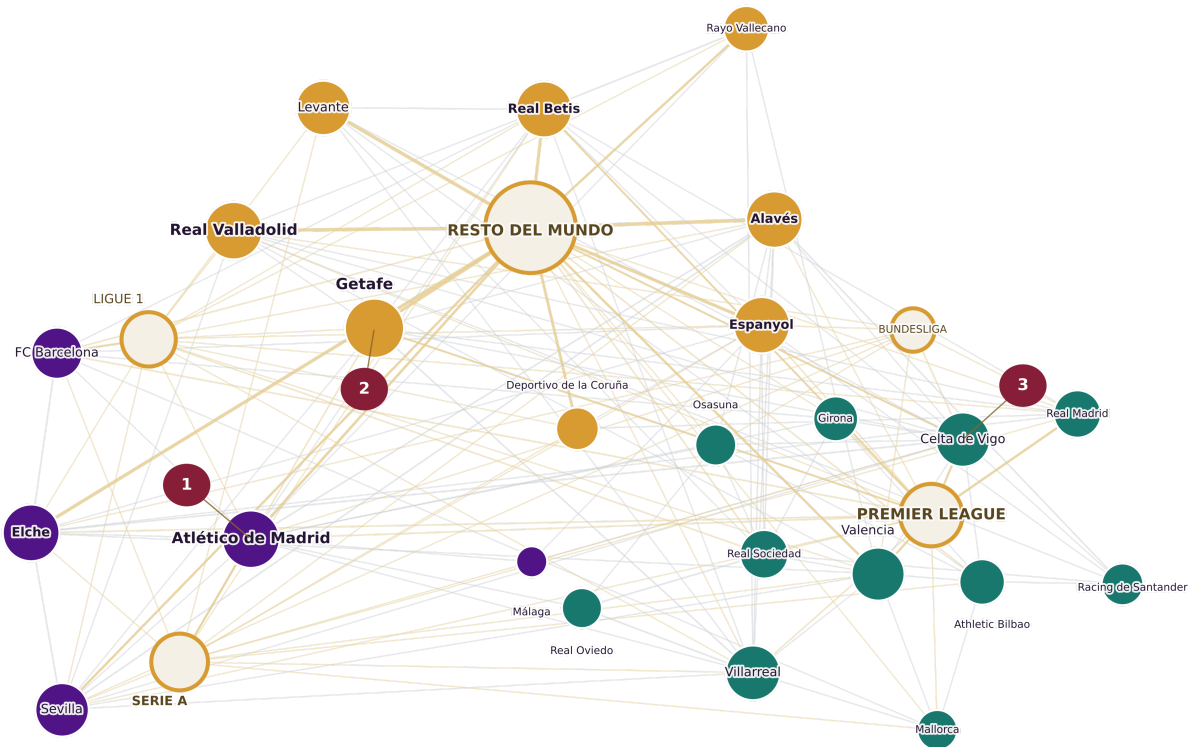


First-Division Club Transfer Network

Which clubs act as hubs and bridges in LaLiga's transfer market, and how does the market split into trading communities?



COMMUNITY (LOUVAIN)

- Group A — Atlético / Barcelona / Sevilla circuit
- Group B — Alavés / Getafe / Betis circuit
- Group C — Real Madrid / Athletic / Valencia circuit

NODE TYPE

- LaLiga First-Division club (color = community)
- Aggregated foreign league / Rest of the World

NODE SIZE & EDGES

- Size $\propto$ weighted degree (players transferred in + out)
- Line width $\propto$ number of players traded between two nodes

Technical Fact Sheet

NETWORK NAME

LaLiga First-Division Transfer Network (2025–2027), filtered from the full UPY Transfer Market Project dataset.

SOURCE

Fichajes.com (official transfer market tracker). Scraped and modeled in the *UpyTransferMarketProject* repository.

FILTERING & METHODS (DECLARED)

Only the 24 current First-Division clubs are kept; reserve/B teams (Alavés B, Real Madrid B, etc.) were removed. Foreign-league transfers stay **aggregated exactly as provided** (Premier League, Serie A, Bundesliga, Ligue 1, Rest of the World) — 5 macro-nodes. Metrics recomputed on this filtered graph with **NetworkX**; communities via **Louvain**.

GRAPH TYPE

Undirected, weighted, simple graph (aggregated from a directed weighted multigraph of individual transfers). Connected — 1 component.

CENTRALITIES APPLIED

Weighted degree (players moved in+out) · **Betweenness** (bridge role) · **PageRank** (influence via transfer flow)

29

NODES (24 CLUBS + 5 LEAGUES)

241

UNIQUE CONNECTIONS

496

PLAYER TRANSFERS

0.40

DENSITY

0.48

AVG. CLUSTERING

3 / 1.62

DIAMETER / AVG. PATH

2–3 FINDINGS (LaLiga clubs only, leagues excluded)

1 Atlético de Madrid is the only club top-3 in all three centralities (degree 42, betweenness 0.068, PageRank 0.044) — hub *and* bridge.

2 Getafe has the highest weighted degree (47) despite linking to only 11 clubs — high-volume, concentrated trading.

3 Celta de Vigo tops betweenness (0.070) though mid-table in degree — a quiet bridge between communities.

TOOLS

Python · BeautifulSoup4 · NetworkX · python-louvain · Matplotlib · HTML/CSS